Week 6 DSC 180 Presentation



DSD Team
Shant, Alyx, Adit

Week 6 Technical Scoping & Advisor Outreach: Define data pipelines; identify & recruit faculty/grad technical advisors (e.g., Time Series).

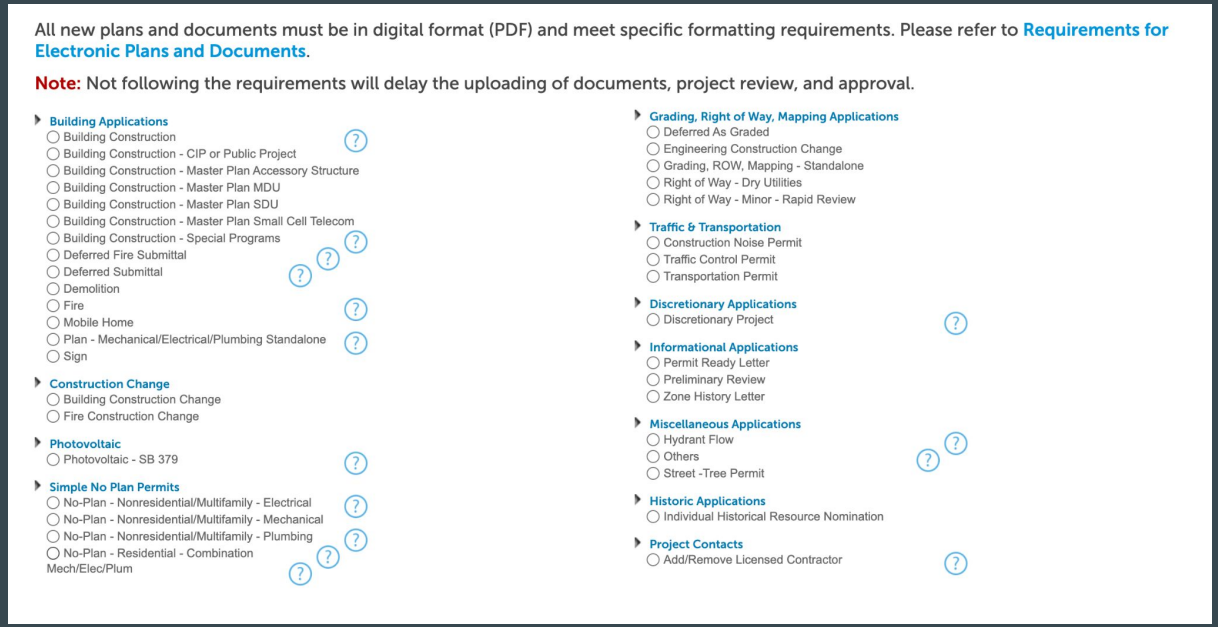
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Improvements to make from last time:

- Less words on slides / add images whenever possible
- Introduce David and Mazen and what they do
- Have less technical info in slides (or take time explaining them if really important)
- Talk slower/shorter presentation
- Briefly explain DSD

Problem Summary / Overview

- **Stakeholder: Developmental Services Department**
  - The City of San Diego's DSD reviews plans and issues permits for construction and development projects.
- Resubmittals slow everyone down ⇒ help applicants get it right the first time.
- Success metric ⇒ lower resubmittal cycle
- Would make permitting process quicker for both applicants and DSD team, benefiting everyone



All new plans and documents must be in digital format (PDF) and meet specific formatting requirements. Please refer to **Requirements for Electronic Plans and Documents**.

Note: Not following the requirements will delay the uploading of documents, project review, and approval.

Building Applications
- Building Construction
- Building Construction - CIP or Public Project
- Building Construction - Master Plan Accessory Structure
- Building Construction - Master Plan MDU
- Building Construction - Master Plan SDU
- Building Construction - Master Plan Small Cell Telecom
- Building Construction - Special Programs
- Deferred Fire Submittal
- Deferred Submittal
- Demolition
- Fire
- Mobile Home
- Plan - Mechanical/Electrical/Plumbing Standalone
- Sign

Construction Change
- Building Construction Change
- Fire Construction Change

Photovoltaic
- Photovoltaic - SB 379

Simple No Plan Permits
- No-Plan - Nonresidential/Multifamily - Electrical
- No-Plan - Nonresidential/Multifamily - Mechanical
- No-Plan - Nonresidential/Multifamily - Plumbing
- No-Plan - Residential - Combination Mech/Elec/Plum

Grading, Right of Way, Mapping Applications
- Deferred As Graded
- Engineering Construction Change
- Grading, ROW, Mapping - Standalone
- Right of Way - Dry Utilities
- Right of Way - Minor - Rapid Review

Traffic & Transportation
- Construction Noise Permit
- Traffic Control Permit
- Transportation Permit

Discretionary Applications
- Discretionary Project

Informational Applications
- Permit Ready Letter
- Preliminary Review
- Zone History Letter

Miscellaneous Applications
- Hydrant Flow
- Others
- Street -Tree Permit

Historic Applications
- Individual Historical Resource Nomination

Project Contacts
- Add/Remove Licensed Contractor

Who's who

- DSD (Development Services Department)
- Apply → Review → Comments → Resubmit → Issued → Inspect.
- David ⇒ Point of contact for domain knowledge, understanding high and ground-level problem
- Mazen ⇒ Point of contact for data extraction, analyses, and brainstorming deliverables



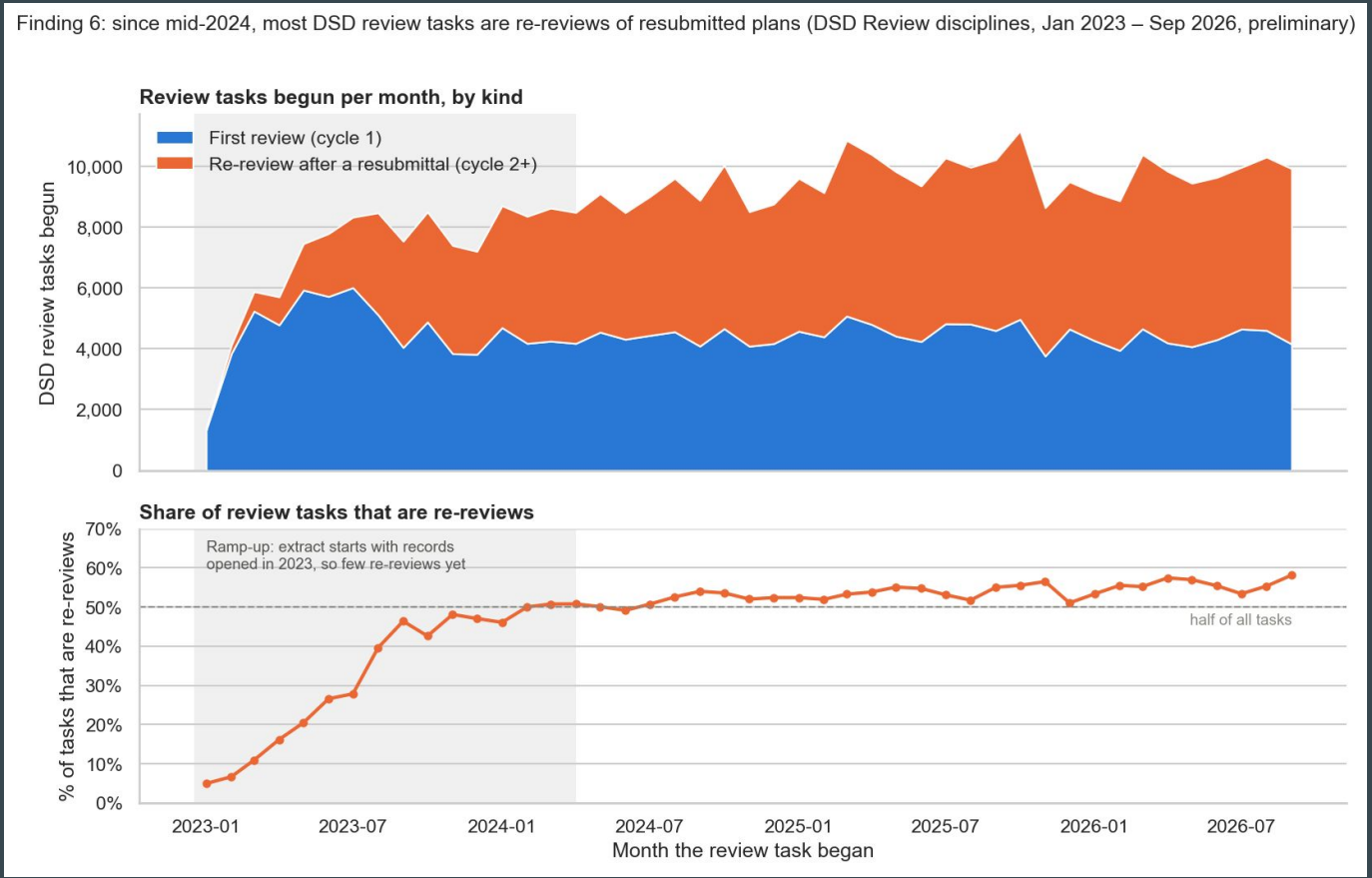
Mazen Abugharbieh
Program Manager at DSD
(Urban Division)



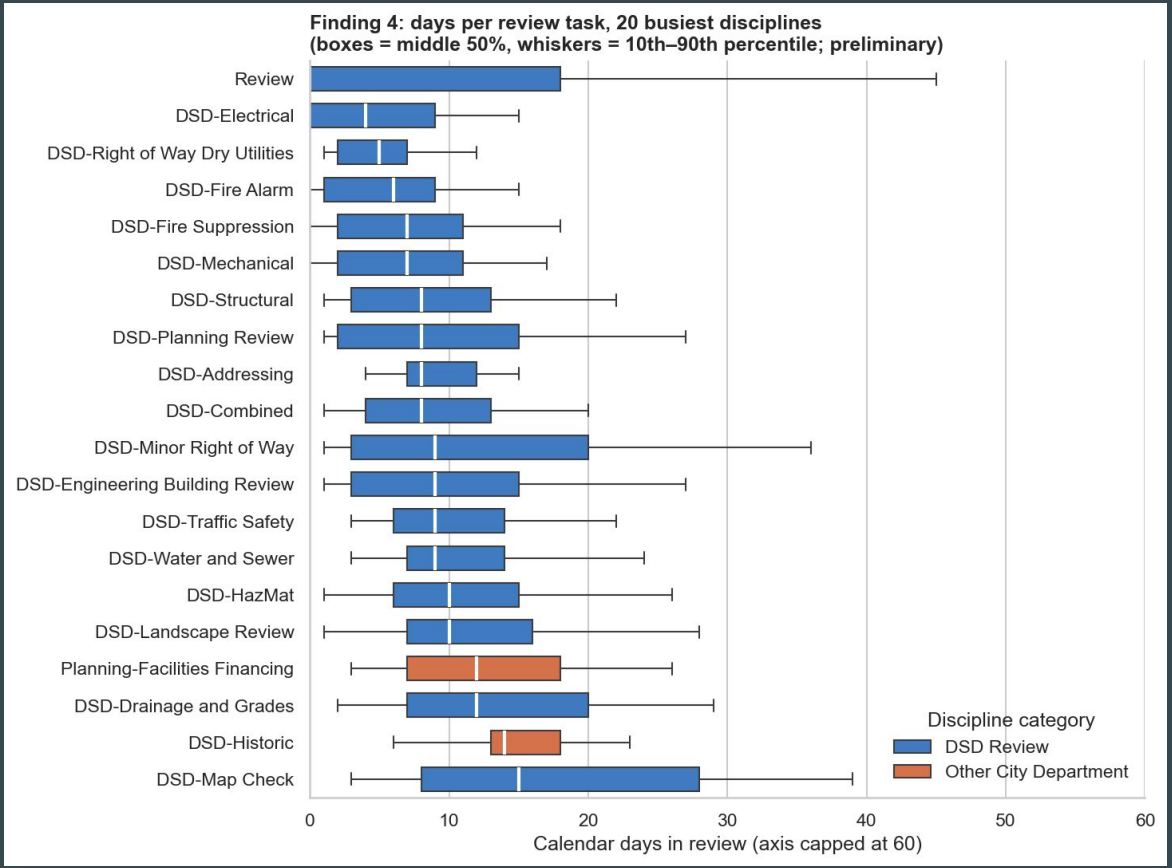
David Saborio
Program Manager at DSD
(Business Operations)

What we learned from the data

- Around 50% of tasks are re-reviews



What we learned from the data



DSD is only as fast as the slowest discipline

Re-scoping the solution

Deliverable A - User Facing Tool

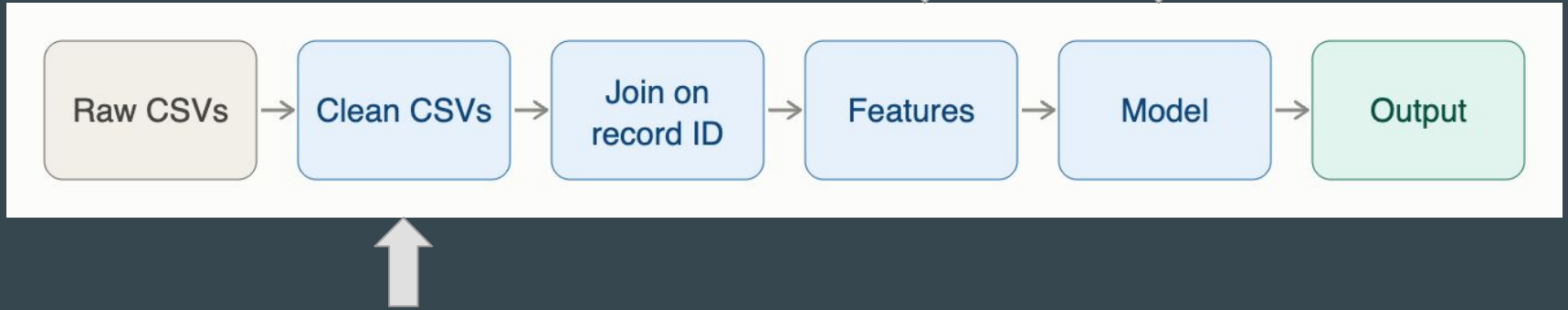
- Predict which permits will need many resubmittals
- Tell applicants what to watch for and guide them to relevant permit codes.

Deliverable B - Operations Analysis

- A time-series view of review workload and capacity
- Some analysis on DSD's operations to identify ways of increasing productivity/efficiency

Technical pipeline

AI Model building steps



Clean some of the data quality issues

Next steps and timeline

Week 6:

- Finalize quick communication method (Slack, Discord, other)
- Meet with David and Mazen to finalize deliverable
- Advisor Outreach → Schedule meeting with permit reviewer*
- Quarter 1 Project Checkpoint due Friday

Moving Forward:

- Keep options open dependent on compute access

*Advisor Outreach ⇒ expressed interest to meet with several permit reviewers to understand pain points from a user point of view.



Thank you!

Open to Questions!